



Department of **Mathematics and Natural Sciences**
MAT120- Mathematics II: Integral Calculus and Differential Equations
Graded Quiz 03
Total marks: 15
Time: 25 minutes

Instructions:

- Please write your name, ID, and section in your answer script.
- Answer all the questions.
- Submit the PDF of your answer script

1. Evaluate $\iint_D (6xy + 4x^2) dA$, where D is the portion of $x^2 + y^2 = 9$ in the second quadrant. [8]

2. Solve the given differential equation by using the separation of variable method,

$$(e^x + e^{-x}) \frac{dy}{dx} = y^2. \quad [7]$$